Technical Introduction

This Java-based Blackjack game was built with a JavaFX GUI. The program follows object-oriented programming principles with different game elements like card, deck, player, and dealer representing different classes as a part of the model. Following an MVC structure, the classes are brought together in a centralized game controller which handles the progression of the game. The view contains the application file which sets up the presentation of the game such as the chip and card images. MVC structure allows for different layers of game development to be fairly modular which is easier to debug and maintain.

User Stories

- As a player, I want to see my cards so I can understand my hand.
- As a player, I want to press “Hit” to draw another card as often as I want or until I bust.
- As a player, I want to press “Stand” to end my turn.
- As a player, I want the game to follow standard Blackjack rules.
- As a player, I want clear notification about the result so I can understand how the game ended.
- As a player, I want to be able to keep track of my winnings or losses so I know how my bets and strategies are progressing.
- As a player, I want the game to continue after a win, loss, or tie so I can play multiple rounds.
- As a player, I want to restart the game when I run out of money so I can continue playing.

Incomplete User Stories

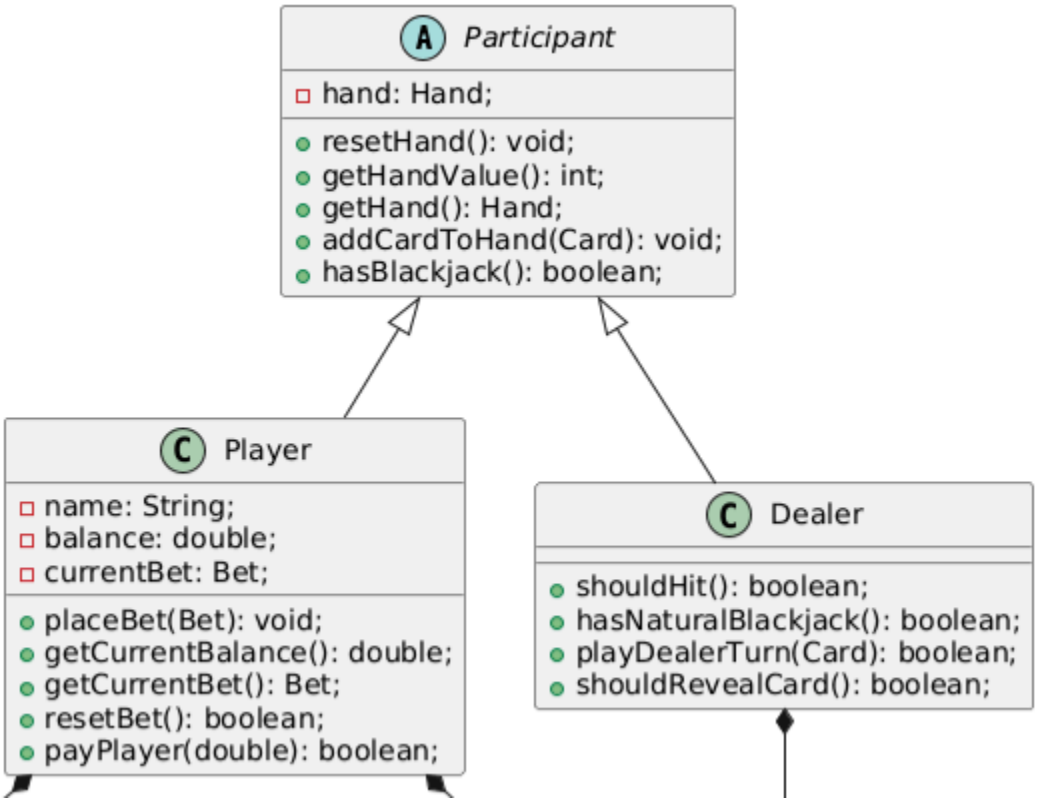
- As a player, I want to split pairs to add a layer of complexity to Blackjack.
- As a player, I want to double down to add a layer of complexity to Blackjack.
- As a player, I want a sound that reads out loud the hand value and game result so the game is playable for people with poor vision.

CRC Cards

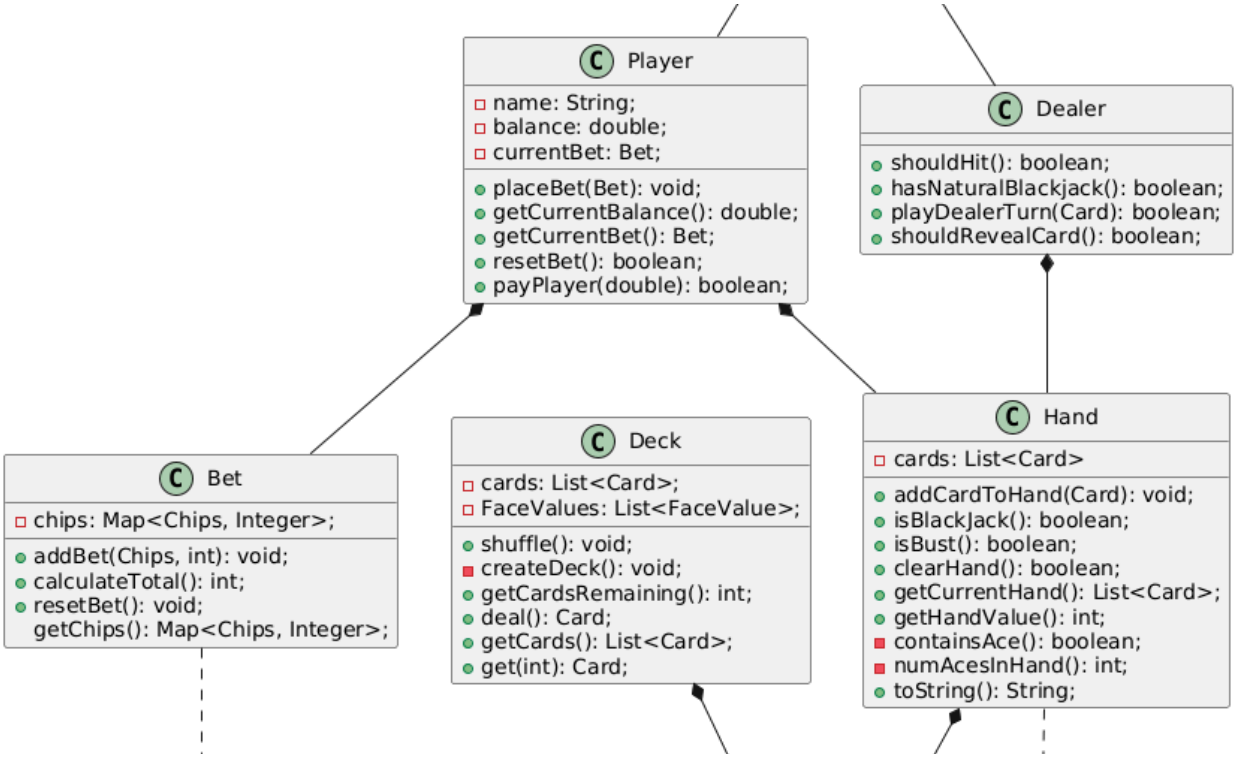


These classes relate the basic classes of objects that are involved in the model of this Blackjack game. The abstract Participant classes are implemented with Player whose actions can be controlled by the person playing the game and Dealer whose actions are controlled by the game logic. The game logic is implemented with the help of several classes like Card which holds all information about each card, the Deck which holds a collection of cards, the Hand which is distributed to both Player and Dealer, and Bet which holds all information about bets placed by the person playing the game.

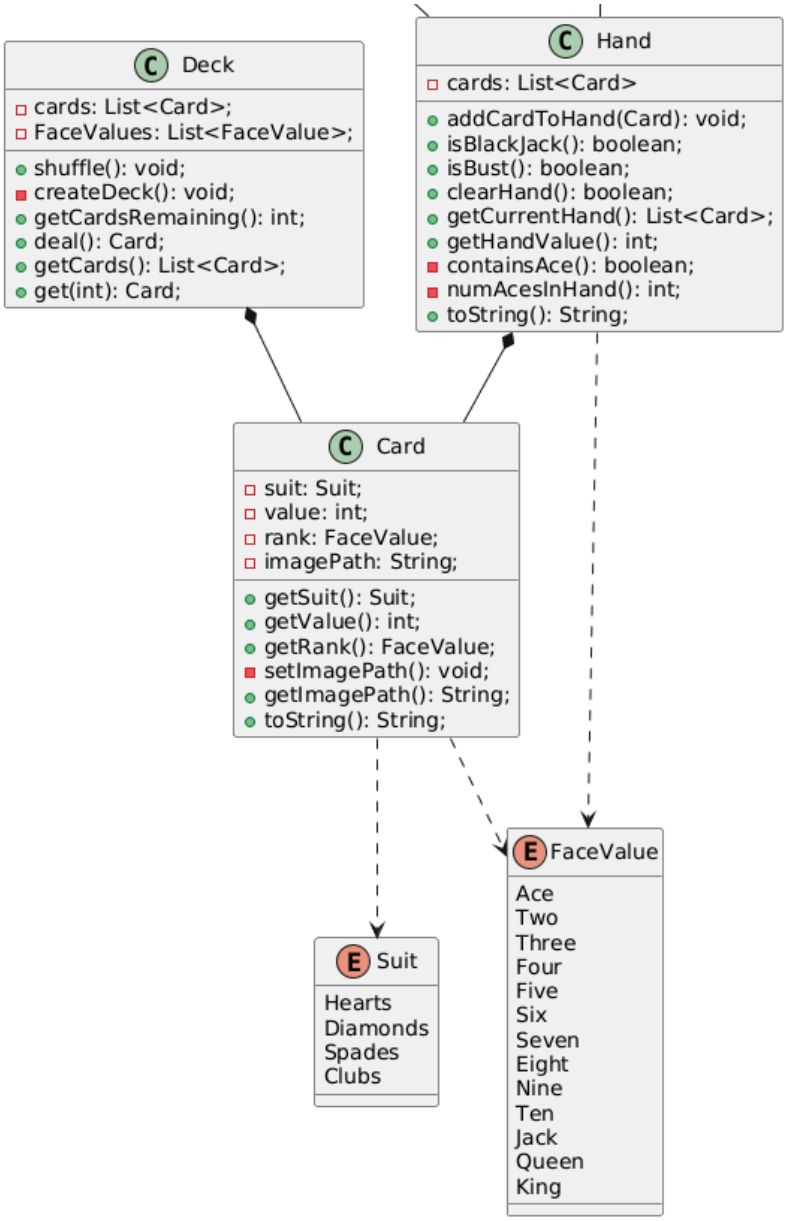
Further Details from UML



As noted, Participant is an abstract class with Player and Dealer as children. Participant was created because every participant in the game must have a hand and basic methods noted in the diagram. Differences are added for each individual child. For example, Player can place a bet (placeBet) but Dealer does not need that method.

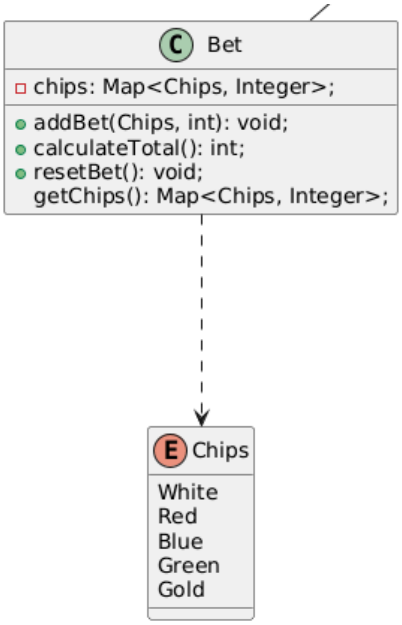


This part of the diagram shows that while both Player and Dealer have access to Hand, only Player can access Bet. While the game should be a joy in itself, winning is also extremely satisfying; players certainly want to know how they are progressing against the dealer. The Bet class makes it simple to keep track of the player's decisions, show the visualization of their bet in chips, easily control/reset their bet, and get the results of the winnings or losses.



Both Deck and Hand hold several cards of the Card class which contain all the attributes that a real card would have like rank and suit as well as getter, string, and image-related methods. Attributes like suit and value are represented by enums which use descriptive and meaningful names and are easier to read and maintain. While Blackjack could be played by description in the terminal, it is more visually engaging and easier on the eyes to make a nice interface with animations mirroring the real game as much as possible. These classes and descriptions make it

possible to generate the right card picture as the game progresses. It is also easier for the player to judge the game and make decisions with clear visuals rather than descriptions of the hand.



Similarly, the Bet class also has an enum with chip types that represent different numerical/monetary values of bet. The colouring scheme mirrors the values of the actual game and are therefore intuitive and easy to understand. Having different coloured chips mirrors a real Blackjack game and will be both more familiar to the player and easier to tell apart bet values even for beginners.